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Scraper for trench closing assembly wheel

Abstract

A trench closing assembly (250) for a row unit (200) of an agricultural planter, comprising a main frame (251) and a frame member (1520) pivotally supported from the main frame, a pair of closing wheels (254) each rotatably supported by the frame member, a mounting arm (1570) pivotally mounted to said frame member and supporting a press wheel (255) rearward of said closing wheels, including a scraper (3600, 4600) adjacent to an inner side of said press wheel.

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Background/Summary

BACKGROUND

(1) It is well known that good seed-to-soil contact within the seed trench is a critical factor in uniform seed emergence and high yields. To obtain good seed-to-soil contact, once seeds are planted in a seed trench, the seed trench needs to be closed so that soil surrounds the seed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side elevation view of an embodiment of a row unit of an agricultural planter.
- (2) FIG. 2 is a diagram of a system for implementing operational control of the closing wheel assembly based on signals generated by the trench closing sensor.
- (3) FIG. 3 is a left side perspective view of an embodiment of a trench closing assembly.
- (4) FIG. 4 is a top plan view of the trench closing assembly of FIG. 3.
- (5) FIG. 5 is a bottom plan view of the trench closing assembly of FIG. 3.
- (6) FIG. 6 is a top perspective view of the trench closing assembly of FIG. 3 with some parts removed.
- (7) FIG. 7 is a bottom perspective view of the trench closing assembly of FIG. 6.
- (8) FIG. 8 is a bottom perspective view of the trench closing assembly of FIG. 7 with some parts removed.
- (9) FIG. 9 is a bottom perspective view of the trench closing assembly of FIG. 8 with some parts removed.
- (10) FIG. 10 is a top perspective view of the trench closing assembly of FIG. 9.
- (11) FIG. 11 is a top perspective view of the frame of the trench closing assembly of FIG. 3.
- (12) FIG. 12 is a bottom perspective view of the frame of FIG. 11.
- (13) FIG. 13 is bottom plan view of the frame of FIG. 11.
- (14) FIG. 14 is top perspective view of the frame of the trench closing assembly of FIG. 3.
- (15) FIG. 15 is a bottom perspective view of the frame of FIG. 14.
- (16) FIG. 16 is a top perspective view of the connection bracket of the frame of FIG. 14.
- (17) FIG. 17 is a bottom perspective view of the connection bracket of FIG. 16.
- (18) FIG. 18 is a top perspective view of the attachment bracket of the frame of FIG. 14.
- (19) FIG. 19 is a bottom perspective view of the attachment bracket of FIG. 15.
- (20) FIG. 20 is top perspective view of the mounting arm of the trench closing assembly of FIG. 3.
- (21) FIG. 21 is a bottom perspective view of the mounting arm of FIG. 20.
- (22) FIG. 22 is a front elevation view of the handle assembly of the trench closing assembly of FIG. 3.
- (23) FIG. 23 is a left side elevation view of the handle assembly of FIG. 22.
- (24) FIG. 24 is a perspective view of the shank of the handle assembly of FIG. 22.
- (25) FIG. 25 is a bottom perspective view of the pressure sensor of the handle assembly of FIG. 22.
- (26) FIG. 26 is a top perspective view of the pressure sensor of FIG. 25.
- (27) FIG. 27 is a perspective view of the handle assembly and mounting arm of the trench closing assembly of FIG. 3.
- (28) FIG. 28 is a perspective view of the trench closing assembly of FIG. 3 further including soil deflectors and a fluid applicator according to an embodiment.

- (29) FIG. 29 is a left side elevation view of the trench closing assembly of FIG. 28.
- (30) FIG. 30 is a top plan view of the trench closing assembly of FIG. 28.
- (31) FIG. 31 is a left side elevation view of the closing wheel soil deflector of FIG. 28.
- (32) FIG. 32 is a left side elevation view of the press wheel soil deflector of FIG. 28.
- (33) FIG. 33 is a perspective view of the closing wheel soil deflector of FIG. 31.
- (34) FIG. 34 is a perspective view of the press wheel soil deflector of FIG. 32.
- (35) FIG. 35 is a top plan view of the closing wheel soil deflector of FIG. 31.
- (36) FIG. 36 is a top plan view of the press wheel soil deflector of FIG. 32.
- (37) FIG. 37 is a perspective view of the fluid applicator according to an embodiment.
- (38) FIG. 38 is a right side elevation view of the fluid applicator of FIG. 37.
- (39) FIG. 39 is a top plan view of the fluid applicator of FIG. 37.
- (40) FIG. 40 is a rear elevation view of the fluid applicator of FIG. 37.
- (41) FIG. 41 is a top plan view of a fluid applicator according to an embodiment.
- (42) FIG. 42 is a left side elevation view of the closing wheel soil deflector of FIG. 31 and including a nozzle.
- (43) FIG. 43 is a left side elevation view of the closing wheel soil deflector of FIG. 32 and including a nozzle.
- (44) FIG. 44 is a perspective view of the trench closing assembly of FIG. 3 with modified closing wheels and scrapers for the press wheels according to an embodiment.
- (45) FIG. 45 is a left side elevation view of the trench closing assembly of FIG. 44.
- (46) FIG. 46 is a left side elevation view of the trench closing assembly of FIG. 45 with the press wheels removed.
- (47) FIG. 47. is a perspective view of the trench closing assembly of FIG. 46.
- (48) FIG. 48 is a perspective view of the scrapers.
- (49) FIG. 49 is a front elevation view of the scrapers of FIG. 48.
- (50) FIG. 50 is a top plan view of the scrapers of FIG. 48.
- (51) FIG. 51 is a bottom plan view of the scrapers of FIG. 48.
- (52) FIG. 52 is a left side elevation view of the left scraper of FIG. 48.
- (53) FIG. 53 is a right side elevation view of the left scraper of FIG. 48.
- (54) FIG. 54 is a perspective view of the left press wheel and left scraper of FIG. 44.
- (55) FIG. 55 is a left side elevation view of the closing wheel of FIG. 44.
- (56) FIG. 56 is a partial perspective view of the trench closing assembly of FIG. 3 with modified scrapers for the press wheels according to an embodiment.
- (57) FIG. 57 is a perspective view of the trench closing assembly of FIG. 56 with the press wheels removed for viewing.
- (58) FIG. 58 is a perspective view of a press wheel and scraper of FIG. 56.
- (59) FIG. 59 is a perspective view of the scrapers of FIG. 56.
- (60) FIG. 60 is a top plan view of the scrapers of FIG. 59.
- (61) FIG. 61 is a bottom plan view of the scrapers of FIG. 59.
- (62) FIG. 62 is a front elevation view of the scrapers of FIG. 59.
- (63) FIG. 63 is a right side elevation view of the scrapers of FIG. 59.
- (64) FIG. 64 is top plan view of the right scraper of FIG. 59.
- (65) FIG. 65 is a bottom plan view of the right scraper of FIG. 64.
- (66) FIG. 66 is a right side elevation view of the right scraper of FIG. 64.
- (67) FIG. 67 is a left side elevation view of the right scraper of FIG. 64.
- (68) FIG. 68 is a front elevation view of the right scraper of FIG. 64.
- (69) FIG. 69 is a rear elevation view of the right scraper of FIG. 64.
- (70) FIG. 70 is a top plan view of the left scraper of FIG. 59.
- (71) FIG. 71 is a bottom plan view of the left scraper of FIG. 70.
- (72) FIG. 72 is a right side elevation view of the left scraper of FIG. 70.

(73) FIG. 73 is a left side elevation view of the left scraper of FIG. 70.

(74) FIG. 74 is a front elevation view of the left scraper of FIG. 70.

(75) FIG. 75 is a rear elevation view of the left scraper of FIG. 70.

DESCRIPTION

(76) All references cited herein are hereby incorporated by reference in their entireties. In the event of a conflict in a definition in the present disclosure and that of a cited reference, the present disclosure controls.

(77) Referring now to the drawings, wherein like reference numerals designate identical or corresponding parts throughout the several views, FIG. 1 illustrates an embodiment of an agricultural planter row unit **200**. The row unit **200** is comprised of a frame **204** pivotally connected to a toolbar **202** by a parallel linkage **206** enabling each row unit **200** to move vertically independently of the toolbar **202**. The frame **204** operably supports one or more hoppers **208**, a seed meter **210**, a seed delivery mechanism **212**, a downforce control system **214**, a seed trench opening assembly **220**, a trench closing assembly **250**, an optional packer wheel assembly **260**, and an optional row cleaner assembly **270**. It should be understood that the row unit **200** shown in FIG. 1 may be for a conventional planter or the row unit **200** may be a central fill planter, in which case the hoppers **208** may be replaced with one or more mini-hoppers and the frame **204** modified accordingly as would be recognized by those of skill in the art.

(78) The optional downforce control system **214** is disposed to apply lift and/or downforce on the row unit **200** such as disclosed in U.S. Publication No. US2014/0090585. The downforce applied by downforce control system **214** can be determined by methods disclosed in U.S. Pat. No. 9,173,339.

(79) The seed trench opening assembly **220** includes a pair of opening discs **222** rotatably supported by a downwardly extending shank member **205** of the frame **204**. The opening discs **222** are arranged to diverge outwardly and rearwardly so as to open a v-shaped trench **10** in the soil **11** as the planter traverses the field. The seed delivery mechanism **212**, such as a seed tube or seed conveyor, is positioned between the opening discs **222** to deliver seed from the seed meter **210** into the opened seed trench **10**. The depth of the seed trench **10** is controlled by a pair of gauge wheels **224** positioned adjacent to the opening discs **222**. The gauge wheels **224** are rotatably supported by gauge wheel arms **226** which are pivotally secured at one end to the frame **204** about pivot pin **228**. A rocker arm **230** is pivotally supported on the frame **204** by a pivot pin **232**. It should be appreciated that rotation of the rocker arm **230** about the pivot pin **232** sets the depth of the trench **10** by limiting the upward travel of the gauge wheel arms **226** (and thus the gauge wheels) relative to the opening discs **222**. The rocker arm **230** may be adjustably positioned via a linear actuator **234** mounted to the row unit frame **204** and pivotally coupled to an upper end of the rocker arm **230**. The linear actuator **234** may be controlled remotely or automatically actuated as disclosed, for example, in International Publication No. WO2014/186810.

(80) An optional downforce sensor **238** is configured to generate a signal related to the amount of force imposed by the gauge wheels **224** on the soil. In some embodiments the pivot pin **232** for the rocker arm **230** may comprise the downforce sensor **238**, such as the instrumented pins disclosed in U.S. Pat. No. 8,561,472.

(81) An optional seed meter **210** may be any commercially available seed meter, such as a finger-type meter or a vacuum seed meter. One exemplary vacuum seed meter is the VSet® meter, available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568.

(82) An optional packer wheel assembly **260** comprises an arm **262** pivotally attached to the row unit frame **204** and extends rearward of the closing wheel assembly **250** and in alignment therewith. The arm **262** rotatably supports a packer wheel **264**. An actuator **266** is pivotally attached at one end to the arm **262** and at its other end to the row unit frame **204** to vary the amount of downforce exerted by the packer wheel **264** to pack the soil over the seed trench **10**.

(83) An optional row cleaner assembly **270** may be the CleanSweep® system available from

Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568. The row cleaner assembly **270** includes an arm **272** pivotally attached to the forward end of the row unit frame **204** and aligned with the trench opening assembly **220**. A pair of row cleaner wheels **274** are rotatably attached to the forward end of the arm **272**. An actuator **276** is pivotally attached at one end to the arm **272** and at its other end to the row unit frame **204** to adjust the downforce on the arm to vary the aggressiveness of the action of the row cleaning wheels **274** depending on the amount of crop residue and soil conditions.

(84) Referring to FIG. 2, a monitor **300** is visible to an operator within the cab of a tractor pulling the planter. The monitor **300** may be in signal communication with a GPS unit **310**, the trench closing assembly actuator **256** and the optional packer wheel assembly actuator **266** to enable operational control of the trench closing assembly **250** and the optional packer wheel assembly **260** based on the signals generated by trench closing sensors **1000**, which are described in International Publication No. WO2017/197274. Also as discussed later, the monitor **300** may be programmed to display operational recommendations based on the signals generated by the trench closing sensors **1000**. The monitor **300** may also be in signal communication with the row cleaner actuator **276**, the downforce control system **214**, the depth adjustment actuator **234** to enable operational control of row cleaner assembly **270**, the downforce control system **214** and the trench opening assembly **230**, respectively.

(85) Trench Closing Assembly

(86) Examples of Trench Closing Assemblies can be found in U.S. Patent Application Nos. 62/637,372, filed 1 Mar. 2018, 62/644,201, filed 16 Mar. 2018, 62/731,813, filed 14 Sep. 2018, 62/791,203, filed on 11 Jan. 2019, 62/831,338, filed 9 Apr. 2019, 62/843,037, filed 3 May 2019, and in International Application No. PCT/US2019/020452, filed on 2 Mar. 2019. Also, an example of a trench closing assembly is the FurrowForce™ trench closing assembly available from Precision Planting, LLC.

(87) FIGS. 3-27 illustrate a trench closing assembly **250D** according to an embodiment. In this embodiment, trench closing assembly **250D** has a main frame **251D** that is connected to row unit **200**. Trench closing assembly **250D** has a frame **1520**, an actuator **259**, a pair of closing wheels **254D-1**, **254D-2**, and optionally, a press wheel **255D**.

(88) Turning to FIGS. 14-19, frame **251D** has a connection bracket **1540** and an attachment bracket **1550**. Connection bracket **1540** can have one or more bolts **1209D** extending through apertures **1207D** in connection bracket **1540** for mounting the frame **251D** to the row unit **200**. Connection bracket **1540** (as with connection brackets **1200** and **1201A**) can be varied to mate attachment to different styles of row units. Connection bracket **1540** has a first side **1542-1**, a second side **1542-2**, and a plate **1543** disposed between first side **1542-1** and second side **1542-2**. First side **1542-1** and second side **1542-2** each have a post **1541-1** and **1541-2**, respectively, protruding perpendicularly outward.

(89) Attachment bracket **1550** connects to connection bracket **1540**. Attachment bracket **1550** provides a common structure for mounting other parts while connection bracket **1540** has a varied structure to mate with different styles of row units. Attachment bracket has a first side **1552-1**, a second side **1552-2**, a crossbar **1557** disposed between first side **1552-1** and second side **1552-2**, and plate **1553** disposed between first side **1552-1** and second side **1552-2**, and plate **1553**. First side **1552-1** and second side **1552-2** each have a u-shaped opening **1555-1** and **1555-2**, respectively, for connection to posts **1541-1** and **1541-2**, respectively. First side **1552-1** and second side **1552-2** have openings **1558-1** and **1558-2**, respectively, for accepting pivots **1529-1** and **1529-2**, respectively. Attachment bracket **1550** can be secured to connection bracket with fastener **1549**. Optionally, attachment bracket **1550** can also have an opening **1501** disposed in first side **1552-1** or second side **1552-2** for accepting a pin. While connection bracket **1540** is illustrated with separate parts it may be fabricated as a unitary part.

(90) Optionally, as illustrated in FIGS. 5 and 7, a guard **1590** can be disposed on frame **1520** ahead

of closing wheels **254D-1**, **254D-2**. Guard **1590** can prevent rocks, rootballs, or other trash from approaching closing wheels **254D-1**, **254D-2**. The height of guard **1590** can be adjusted by changing the placement of arms **1591-1** and **1591-2** on frame **1520**.

(91) Turning to FIGS. **20-21**, mounting arm **1570** has a first side **1571-1**, a second side **1571-2**, a plate **1574** disposed between first side **1571-1** and second side **1571-2**, a hole **1573** through plate **1574** for passage of sleeve **1603**, axle hubs **1575-1** and **1575-2** for mounting press wheels **255D-1** and **255D-2**, respectively, and, optionally, a handle cradle **1579**.

(92) Turning to FIGS. **6-13**, frame **1520** is pivotally connected to frame **251D** through pivots **1529-1** and **1529-2**. Frame **1520** has a first side **1526-1** and a second side **1526-2**, a plate **1525** connecting first side **1526-1** and second side **1526-2**, a connection arm **1527** extending rearwardly along the direction of travel from plate **1525**, a cross-brace **1528-1** connecting connection arm **1527** to first arm **1526-1**, a cross-brace **1528-2** connecting connection arm **1527** to second arm **1526-2**. First side **1526-1** and second side **1526-2** have openings **1521-1** and **1521-2** for disposing about pivots **1529-1** and **1529-2**, respectively. Mounting arm **1570** can attach to frame **1520** at connections **1522-1a** and **1522-2a** or to **1522-1b** and **1522-2b**. The plurality of connections allows the distance between closing wheels **254D-1**, **254D-2**, and press wheel **255D** (**255D-1**, **255D-2**) to be changed. There can be one connection **1522** or a plurality of connections **1522**. As the mounting arm **1570** position is changed, there are also corresponding connections **1523-a** and **1523-b** (matching in number to connection **1522**) for connecting handle assembly **1600** to frame **1520**. Frame **1520** also has connections **1524** (**1524-1a**, **1524-1b**, **1524-2a**, and **1524-2b**) for mounting closing wheels **254D-1**, **254D-2**. While there can be one set of connections **1524**, the plurality of connections **1524** allow for forward and back placement of the closing wheels **254D-1**, **254D-2** on frame **1520**, or closing wheels **254D-1**, **254D-2** can be offset from each other with one closing wheels **254D-1**, **254D-2** being mounted to a forward location (the “a” position) or to a rear location (the “b” position). As illustrated, closing wheels **254D-1**, **254D-2** are offset from each other. Optionally, frame **1520** can have an opening **1502** in either the first side **1526-1** or second side **1526-2** (shown in **1526-2**) for accepting a pin. Frame **1520** can be raised to allow opening **1501** and opening **1502** to align for accepting a pin (not shown). This allows trench closing assembly **250D** to be raised for transport or when closing is not needed. While illustrated on one side, openings **1501** and **1502** can be disposed on both sides.

(93) Actuator **259** is disposed between plate **1525** and plate **1553** to apply a force to plate **1553** to cause frame **1520** to pivot and apply pressure to closing wheels **254D-1** and **254D-2**.

(94) Turning to FIGS. **22-27**, handle assembly **1600** is illustrated. Handle assembly **1600** has a sleeve **1603** having a first diameter **1620** and a second diameter **1621**. Second diameter **1621** is small enough to be disposed through load sensor **1610**, and first diameter **1620** is large enough so that it cannot pass through load sensor **1610**. Sleeve **1603** has a bracket **1622** (u-shaped bracket as illustrated or any other shape) for accepting handle **1609**. Disposed on sleeve **1603** below bracket **1622** are bevel washers **1601** and **1602**. Bevel washer **1601** and bevel washer **1602** are disposed with their concave surfaces facing each other. This allows bevel washers **1601** and **1602** to flex to absorb shocks experienced by trench closing assembly **250D** to prevent overloading of load sensor **1610**. Before bevel washers **1601** and **1602** reach maximum flex, first diameter **1620** will contact plate **1604** to limit the travel. Load sensor **1610** is a pancake load sensor. Load sensor **1610** has a hole **1612** for passage of sleeve **1603**. Disposed on the underside of load sensor **1610** are a plurality of feet **1611** to allow load sensor **1610** to flex and measure force. Load sensor **1610** can be disposed on mounting arm **1570** directly, or as shown, a plate **1604** can be disposed between load sensor **1610** and mounting arm **1570**. Optionally, a washer **1607** can be disposed about sleeve **1603** under mounting arm **1570**. Handle assembly **1600** is connected to frame **1520** at connection **1523-a** or **1523-b** with a bracket **1606** (u-shaped bracket) and a bolt **1605** connecting bracket **1606** with sleeve **1603**. In the horizontal position, handle **1609** locks handle assembly **1600** in place against mounting arm **1570**. In the vertical position, handle **1609** releases handle assembly from

engagement with mounting arm **1570**. Bolt **1605** can be adjusted to set a vertical placement of mounting arm **1570** relative to frame **1520**.

(95) Load sensor **1610** can be connected to a network directly through a plug (not shown) having a CAN processor to allow direct communication over a CAN network. The CAN processor can communicate pressure readings and provide control signals over the CAN network. Alternatively, load sensor **1610** can be connected to a control module (either an on-row module, or a module controlling a plurality of row) to communicate pressure readings that are then processed by the control module.

(96) Soil Deflector

(97) FIGS. **28-30** illustrate soil deflectors (**3100**, **3500**) and fluid applicator (**2000**) disposed on closing assembly **250D**. Soil deflector **3100** is illustrated on the left, and the right (not shown) is the mirror image of the left. Soil deflectors **3100** and soil deflectors **3500** are illustrated as two parts, but they can be made unitary parts. Soil deflectors **3100** and **3500** can keep dirt/mud from spraying or “rooster tailing” as the trench closing system **250D** traverses the field. Also, soil deflectors **3100** and **3500** can scrape closing wheels **254D** and/or press wheels **255D**, respectively, by being adjusted to contact closing wheels **254D** and/or press wheels **255D**.

(98) Soil deflector **3100-2** is connected to axle **257-2** and extends rearward opposite the direction of travel adjacent to closing wheel **254D-2**. Depending on the shape of closing wheel **254D**, soil deflector **3100** can be shaped to follow the contour of closing wheel **254D** or contact closing wheel **254D**. Soil deflector **3100-2** can be made of two parts, bracket **3101-2** and deflector **3102-2**. When connected to axle **257-2**, optionally a spacer (not shown), such as washer or bushing, can be disposed on axle **257-2** to vary the spacing between soil deflector **3100-2** and closing wheel **254D-2**. Optionally, a connection orientation between bracket **3102-2** and deflector **3102-2** can be changed to change the orientation of deflector **3102-2** with closing wheel **254D-2**. In one embodiment, soil deflector **3100** extends beyond the circumference of closing wheel **254D**, and optionally, extends behind closing wheel **254D** toward frame **1520**. Deflector **3102-2** can be made from a flexible material. A flexible material is a material that will deflect when impacted by soil when trench closing assembly **250D** is operated at typical planting speeds (e.g., up to 20 mph (32 kph) or up to 10 mph (16 kph)). In one embodiment, deflector **3102-2** is polyurethane.

(99) Optionally, soil deflector **3500** (**3500-1**, **3500-2**) can be included for press wheels **255D** (**255D-1**, **255D-2**). While illustrated with two press wheels **255D** (**255D-1**, **255D-2**), there can be just one press wheel **255D**. Soil deflector **3500** (**3500-1**, **3500-2**) is connected to axle **258** (**258-1**, **258-2**) and extends rearward opposite the direction of travel adjacent to press wheel **255D** (**255D-1**, **255D-2**). Depending on the shape of press wheel **255D**, soil deflector **3500** can be shaped to follow the contour of press wheel **255D** or press wheel **255D**. Soil deflector **3500-2** can be made of two parts, bracket **3501-2** and deflector **3502-2**. When connected to axle **258-2**, optionally a spacer (not shown), such as washer or bushing, can be disposed on axle **258-2** to vary the spacing between soil deflector **3500-2** and closing wheel **2545-2**. Optionally, a connection orientation between bracket **3502-2** and deflector **3502-2** can be changed to change the orientation of deflector **3502-2** with press wheel **255D-2**. In one embodiment, soil deflector **3500** extend beyond the circumference of press wheel **255D**, and optionally, extend behind press wheel **255D**. Deflector **3502-2** can be made from a flexible material as defined above.

(100) Fluid Applicator

(101) Fluid applicator **2000** is illustrated in FIGS. **37-40** according to one embodiment. Fluid applicator **2000** has a frame **2100** for connection to mounting arm **1570**. An optional slot **2101** can be disposed in frame **2100** to allow for adjustment front to back along the direction of travel. While illustrated with two arms **2104** (**2104-1**, **2104-2**), fluid applicator **2000** can have just one arm **2014**. Arms **2104** (**2104-1**, **2104-2**) are disposed on either side of mounting arm **1570**. Nozzles **2120** (**2120-1**, **2120-2**) are attached to arms **2104** (**2104**, **1**, **2014-2**), respectively. Optionally, arms **2104** (**2104-1**, **2104-2**) can have slots **2105** (**2105-1-a**, **2105-1-b**, **2105-2-a**, **2105-2-b**) to allows nozzles

2120 (2120-1, 2120-2) to be adjusted vertically. For adjustable connection, U-bolts 2110 (2110-1, 2110-2) can connect nozzle 2120 (2120-1, 2120-2) to arms 2104 (2104-1, 2104-2) through slots 2105 (2105-1-a, 2105-1-b, 2105-2-a, 2105-2-b). Optionally, slots 2105 (2105-1-a, 2105-1-b, 2105-2-a, 2105-2-b) can be sized larger than U-bolts 2110 (2110-1, 2110-2) to allow nozzles 2120 (2120-1, 2120-2) to rotate in a direction transverse to the direction of travel so that nozzles 2120 (2120-1, 2120-2) can deposit fluid closer to or further away from the trench. Optionally, frames 2111 (2111-1, 2111-2) can be included to ensure fit of nozzles 2120 (2120-1, 2120-2) to arms 2104 (2104-1, 2104-2), for example when nozzles 2120 (2120-1, 2120-2) are round. Nozzles 2120 (2120-1, 2120-2) are connected to a fluid source (not shown) for applying fluid to the soil surface. Optionally, frame 2100 can be disposed on mounting arm 1570 rotated 180° from the orientation illustrated.

(102) In an alternative embodiment, fluid applicator 2000' is illustrated in FIG. 41. While illustrated on the right side, fluid applicator 2000' can also be installed as a mirror image on the left side. An arm 2200-1 is connected to an axle 257-1 between frame 1520 and closing wheel 254D-1 and extends in a rearward direction away from the direction of travel. A nozzle 2220-1 is attached to arm 2200-1 via a bracket 2210-1. Optionally, a slot 2205-1 can be disposed in arm 2200-1 to permit adjustment of nozzle 2220-1 front to back along the direction of travel. Nozzle 2220-1 is connected to a fluid source (not shown) for applying fluid to the soil surface.

(103) In an alternative embodiment as illustrated in FIG. 42 or FIG. 43, a nozzle 2300 (2300-2) can be connected to one or both soil deflectors 3100, 3500. Nozzle 2300 is connected to a fluid source (not shown). While illustrated with nozzle 2300 depositing fluid rearward of soil deflector 3100 or 3500 opposite the direction of travel, nozzle 2300 can be positioned to apply fluid towards the trench, or nozzle 2300 can be positioned to deposit fluid away from the trench.

(104) FIGS. 44 to 54 illustrate a scraper 3600 (3600-1, 3600-2) for press wheel 255. As illustrated in these figures, press wheel 255E is the press wheel described in U.S. Application No. 62/843,037, filed 3 May 2019. While illustrated with this press wheel 255E, scraper 3600 can be used with any of the other press wheels 255 described above.

(105) As seen in FIG. 47, scraper 3600 (3600-1, 3600-2) is connected to axle 258 (258-1, 258-2). In an alternative embodiment, scraper 3600 (3600-1, 3600-2) can be connected to axle hub 1575 (1575-1, 1575-2) or mounting arm 1570 (not shown). As seen in FIG. 54, scraper 3600 (3600-1, 3600-2) is disposed at inner side 3650 (3650-1, 3650-2) of wheel 255E (255E-1, 255E-2). Scraper 3600 (3600-1, 3600-2) has a base 3601 (3601-1, 3601-2) which is connected to axle 258 (258-1, 258-2), axle hub 1575 (1575-1, 1575-2), or mounting arm 1570 as described above. Connected to base 3601 3601 (3601-1, 3601-2) is a first end 3602 (3602-1, 3602-2) that extends to a second end 3603 (3603-1, 3603-2). In one embodiment, first end 3602 (3602-1, 3602-2) is wider than second end 3603 (3603-1, 3603-2) such that there is a taper from first end 3602 (3602-1, 3602-2) that extends to a second end 3603 (3603-1, 3603-2). In one embodiment, scraper 3600 (3600-1, 3600-2) is oriented such that second end 3603 (3603-1, 3603-2) is further from the seed trench as compared to first end 3602 (3602-1, 3602-2).

(106) As viewed in FIG. 51, there is an angle α (α -1, α -2) between base 3601 (3601-1, 3601-2) and first end 3602 (3602-1, 3602-2). In some embodiments, the angle is from 90° to 180°. In other embodiments, the angle is 120°-175°, 140°-170°, 150°-160°, or about 155°.

(107) FIG. 55 illustrates closing wheel 254E-2. While illustrated for closing wheel 254E-2, the same can apply to closing wheel 254E-1. Closing wheel 254 can have one or more depth indicators 3700 (3700-a, 3700-b). Depth indicators 3700 indicate the depth that closing wheel 254E has penetrated the soil. This can provide a quick visual indication of the depth by looking at depth indicator 3700. Depth indicator 3700 may or may not include markings for the actual depth associated with depth indicator 3700. There can be any number of depth indicators 3700. The number of depth indicators 3700 can be limited to a number for different selected depths. For example, if typical depth will be 1.5" or 1.75", there will be just two depth indicators 3700. Depth indicators 3700 can be disposed as a circle around closing wheel 254. The placement of depth

indicator **3700** is based on a vertical distance that is perpendicular to the plane that the furthest point on closing wheel **254E** touches on the plane. In the illustrated embodiments, closing wheel **254E** is curved. The placement of depth indicator **3700** on closing wheel **254E** accounts for the curve as is it is placed as described previously.

(108) FIGS. **56** to **75** illustrate a scraper **4600** (**4600-1**, **4600-2**) for press wheel **255E** (**255E-1**, **255E-2**). Scraper **4600** can be used with any of the press wheels **255** described above.

(109) As seen in FIGS. **56-57**, scraper **4600** (**4600-1**, **4600-2**) is connected to axle **258** (**258-1**, **258-2**). In an alternative embodiment, scraper **4600** (**4600-1**, **4600-2**) can be connected to axle hub **1575** (**1575-1**, **1575-2**) or mounting arm **1570** (not shown). As seen in FIG. **54**, scraper **4600** (**4600-1**, **4600-2**) is disposed at inner side **3650** (**3650-1**, **3650-2**) of wheel **255E** (**255E-1**, **255E-2**). Scraper **4600** (**4600-1**, **4600-2**) has a base **4601** (**4601-1**, **4601-2**) which is connected to axle **258** (**258-1**, **258-2**), axle hub **1575** (**1575-1**, **1575-2**), or mounting arm **1570** as described above. Connected to base **4601** **4601** (**4601-1**, **4601-2**) is a first end **4602** (**4602-1**, **4602-2**) that extends to a second end **4603** (**4603-1**, **4603-2**). In one embodiment, first end **4602** (**4602-1**, **4602-2**) is wider than second end **4603** (**4603-1**, **4603-2**) such that there is a taper from first end **4602** (**4602-1**, **4602-2**) that extends to a second end **4603** (**4603-1**, **4603-2**). In one embodiment, scraper **4600** (**4600-1**, **4600-2**) is oriented such that second end **4603** (**4603-1**, **4603-2**) is further from the seed trench as compared to first end **4602** (**4602-1**, **4602-2**). In another embodiment, first end **4602** (**4602-1**, **4602-2**) is generally perpendicular to base **4601** (**4601-1**, **4601-2**) having an angle of 80° to 100° or about 90° . In one embodiment, one scraper **4600-1** can be disposed with respect to the other scraper **4600-2** such that first end **4602-1** of right scraper **4600-1** is disposed vertically above and at least partially overlaps first end **4602-2** of left scraper **4600-2**. This allows both scrapers **4600-1**, **4660-2** to fit between press wheels **255-1**, **255-2**. Second end **4603** (**4603-1**, **4603-2**) can be shaped to conform to inner side **3650** (**3650-1**, **3650-2**) of press wheel **255E** (**255E-1**, **255E-2**).

EXAMPLES

(110) The following are non-limiting examples of various embodiments.

(111) Example 1—A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, said press wheel having an inner side disposed proximate to said frame member; a scraper disposed adjacent to said inner side of said press wheel.

(112) Example 2.—the trench closing system of Example 1, wherein said scraper is connect to an axle that connects said press wheel to said frame member.

(113) Example 3—the trench closing system of Example 1, wherein said scraper has a base for connecting to said frame member, a first end connected to said base, and a second end extending from said first end.

(114) Example 4—the trench closing system of Example 3, wherein said first end is wider than said second end.

(115) Example 5—the trench closing system of Example 3, wherein an angle between said base and said first end is 90° to 180° .

(116) Example 6—the trench closing system of Example 3, wherein an angle between said base and said first end is 120° - 175° .

(117) Example 7—the trench closing system of Example 3, wherein an angle between said base

and said first end is 140°-170°.

(118) Example 8—the trench closing system of Example 3, wherein an angle between said base and said first end is 150°-160°.

(119) Example 9—the trench closing system of Example 3, wherein an angle between said base and said first end is about 155°.

(120) Example 10—the trench closing system of any of Examples 3 to 9, wherein said scraper is disposed such that said second end is further from said seed trench than said first end.

(121) Example 11—the trench closing system of any of any preceding Example further comprising a second press wheel supported by said mounting arm and a second scraper disposed adjacent to an inner side of said second press wheel.

(122) Example 12—the trench closing system of Example 11, wherein the scraper is disposed above and partially overlaps the second scraper.

(123) Various embodiments of the invention have been described above for purposes of illustrating the details thereof and to enable one of ordinary skill in the art to make and use the invention. The details and features of the disclosed embodiments are not intended to be limiting, as many variations and modifications will be readily apparent to those of skill in the art.

Claims

1. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, said press wheel having an inner side disposed proximate to said frame member; a scraper disposed adjacent to said inner side of said press wheel; wherein said scraper has a base for connecting to said frame member, a first end connected to said base, and a second end extending from said first end; and wherein said scraper is disposed such that said second end is further from said seed trench than said first end.

2. The trench closing system of claim 1, wherein said scraper is connected to an axle that connects said press wheel to said frame member.

3. The trench closing system of claim 1, wherein said scraper has a base for connecting to said frame member, a first end connected to said base, and a second end extending from said first end.

4. The trench closing system of claim 3, wherein said first end is wider than said second end.

5. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, said press wheel having an inner side disposed proximate to said frame member; a scraper disposed adjacent to said inner side of said press wheel; wherein said scraper has a base for connecting to said frame member, a first

end connected to said base, and a second end extending from said first end; and wherein an angle between said base and said first end is 120° - 175° .

6. The trench closing system of claim 5, wherein an angle between said base and said first end is 140° - 170° .

7. The trench closing system of claim 5, wherein an angle between said base and said first end is 150° - 160° .

8. The trench closing system of claim 5, wherein an angle between said base and said first end is about 155° .

9. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, said press wheel having an inner side disposed proximate to said frame member; a scraper disposed adjacent to said inner side of said press wheel; a second press wheel supported by said mounting arm and a second scraper disposed adjacent to an inner side of said second press wheel; and wherein the scraper is disposed above and at least partially overlaps the second scraper.
